

Syed Ali Hassan Jaffri

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Profile

Final-year Computer Science student working on the reliability of retrieval-augmented and agentic LLM systems. My final-year project studies how retrieval corpora and conversational memory in agentic RAG pipelines can be poisoned so that a model becomes confidently wrong without ever visibly failing, and builds a deployable defense layer that screens memory and filters poisoned context before either reaches the model. I am drawn to research that treats data, and how it moves through a pipeline, as the primary lever for trustworthy AI rather than architecture alone, and I am looking for research opportunities that let me pursue this domain.

Research Interests

- **Data-centric trust in RAG and agentic systems** – how poisoned or adversarial data entering a pipeline, whether through the retrieval corpus or conversational memory, propagates into downstream model behavior, and how to intervene on the data path rather than the model itself.
- **Traceable retrieval and generation** – designing retrieval and generation systems whose outputs can be traced back to, and checked against, the evidence they were built from, so a wrong answer is catchable rather than just plausible-sounding.
- **Robustness of LLM-based evaluators** – benchmarking where LLM-based detectors (for vulnerabilities, poisoned content, etc.) silently fail against adversarially transformed input, and why.

Education

National University of Sciences and Technology (NUST), SEECS

Islamabad, Pakistan

B.S. in Computer Science

Expected 2027

Dean's Honor List (twice)

Army Public School (APS)

Pakistan

Matriculation; 4th position, country-wide

2021

Final-Year Project

Detecting Memory and Corpus Poisoning in Agentic RAG Pipelines

2026 – Present

- Studying two failure modes in agentic RAG systems: adversarial documents planted in the retrieval corpus, and false memories implanted through ordinary-looking conversation turns – attacks that degrade correctness without producing any visible error or crash.
- Synthesizing ideas from MEMSAD, RAGuard, and TrustRAG into a single deployable layer that screens conversational memory before it is trusted and filters poisoned documents during retrieval, without modifying or retraining the underlying model.
- Designing the evaluation protocol around attack success rate versus benign-query utility, so that a defense is only counted as working if it does not degrade normal use.

Related side projects, built independently to prototype and measure these ideas:

- **RAGShield** – Built an indirect prompt-injection / corpus-poisoning attack against a RAG pipeline, then a defense against it, and measured the gap directly: attack success rate fell from 0.667 to 0.333 on Llama 3.1 8B while benign-query utility was held constant. (github.com/Ali-Sherazii/RagShield)

- **AegisVault** – An ML-based web application firewall built around a layered decision pipeline; reached 99.6% classification accuracy and brought false positives down from 23% to 3.2% over the course of evaluation. (github.com/Ali-Sherazii/AegisVault)

Publications

SoK: Code Obfuscation in the Age of LLM-based Vulnerability Detection

In preparation, 2026

- Systematizes how cheap, semantics-preserving obfuscation collapses both LLM-based and traditional static-analysis vulnerability detectors.
- Builds a benchmark pipeline integrating CodeQL, Infer, Cppcheck, clang-tidy, Snyk, and Coverity as non-LLM baselines, to separate genuine detection capability from pattern-matching that breaks under transformation.

Quantum Threats to the IPFS: A Post-Quantum Migration Framework

Under review, IEEE FIT 2026

- Co-authored study examining post-quantum security implications for content-addressed storage on IPFS.
- Designed a migration strategy leveraging hybrid key exchange (classical + post-quantum) to ensure backward compatibility and quantum resilience.

Research & Engineering Experience

Research Assistant

Apr 2026 – Present

SEECs, NUST

- Benchmarking how LLM-based vulnerability detectors hold up against traditional SAST tools once source code has been obfuscated, feeding directly into the SoK paper above.

LLM Tooling Engineer (Part-time)

Aug 2026 – Present, Remote

Dataquartz

- Working on code obfuscation and IP-protection tooling, and LLM-assisted static analysis workflows.

Blockchain Engineering Intern

Jun – Sep 2026, Remote

Özyeğin University, under Dr. Mesut Sayın

- Built the blockchain integration layer – five role-based Solidity smart contracts and a Python/FastAPI backend – for a TÜBİTAK-funded adaptive vehicle-routing project.
- Outside my core research focus, but it sharpened how I reason about verifiable, auditable system design under adversarial and multi-party assumptions.

Data Science Intern

Jul 2026

10Pearls

- Built an end-to-end serverless pipeline forecasting a city's Air Quality Index three days ahead, with a feature store, CI/CD, and SHAP-based explanations for model predictions.

Skills

ML / NLP

Python, PyTorch, LangChain, Chroma, Ollama, scikit-learn, embeddings, SHAP

Security

Prompt-injection defense, adversarial ML evaluation, SAST (CodeQL, Infer, Snyk), threat modeling

Systems

FastAPI, Docker, GitHub Actions, AWS, Azure

Other

Solidity / web3, MIP / ALNS solvers, SQL, C/C++, JavaScript / TypeScript